Machine Learning Project - Diploma Level of BS in Data Science and Applications

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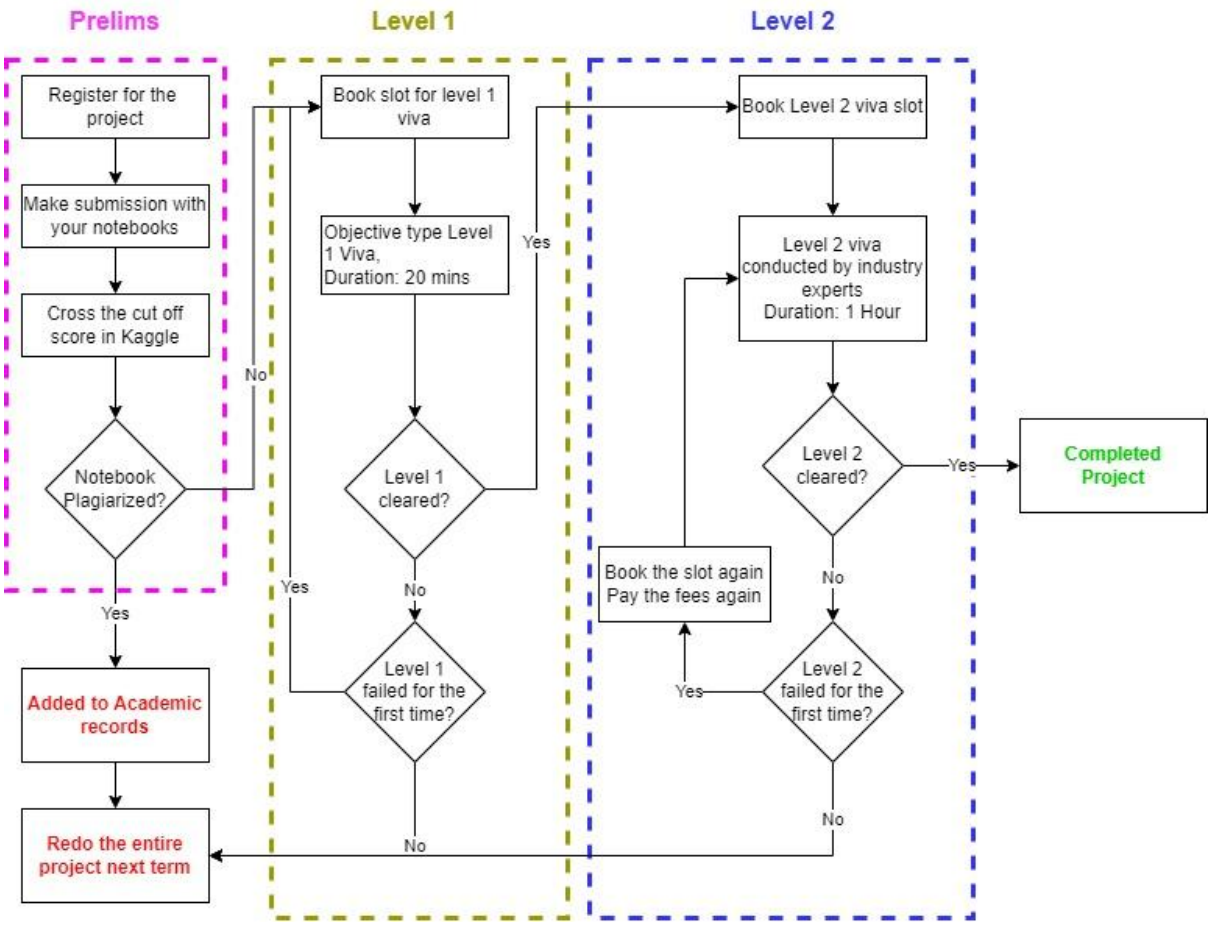
Project Registration Easy 7 Steps

0. Since you will be participating individually, crea**te** a Kaggle account if you don't already have one.

1. Join the Competition using the following link: (https://www.kaggle.com/t/29651a586efd4d448941ba6fa5561e32)
2. After joining the competition, go to the **code tab** and create a **New Notebook**.
3. Keep notebook name as **YourRollNo-notebook-t12026** . (i.e 21f1001234-notebook-t12026)
4. Share the notebook with **iitmbscs2008p** collaborator. View access is sufficient and make/keep it **private**.
5. Build your first ML model (dummy or any other) and make your first submission.
6. **Submit** your **notebook link** in the Competition Registration Form.

Without doing this step, which is submitting your Kaggle Notebook link in the Competition Registration Form, you will not be eligible to complete the project or do the viva.

Project Workflow



Duration of the project

- **Release of project statement: Jan 17, 2026**
- **Extension of Deadline for submitting notebook link: Feb 28, 2026**
- **Kaggle competition deadline : 15-03-2025**
- **Latest Viva & Evaluation Policy: click here**
- **MLP Project Evaluation Timelines and Scheme:**

MLP Course theory completion Term	**Deadline to Cross Cutoff**	**Deadline to Complete Both Vivas**	**Evaluation Scheme**
May 2025 and earlier	15 March 2026	30 March 2026	Evaluated out of 100 marks
Sep 2025 and Jan 2026	15 March 2026	30 March 2026	Evaluated out of 105 marks, with the final score capped at 100

The deadline to cross cut-off is extended to 15 March 2026 for students from May 2025 and earlier terms. Please note that the **deadline will not be extended beyond 15 March 2026**.

Note: You may continue to participate in the competition and **improve your score even after completing your viva**, until the competition ends. However, students **who do not meet the deadline for completing both vivas will not be given any further opportunity to appear for the vivas later.**

Criteria for bonus marks(As per grading doc):

Marking for the projects w.e.f t12026 :

If you complete the project in the same term as the theory course or the subsequent term, we will evaluate your project for 105 marks and cap it at 100. Thereafter, the projects will be evaluated out of 100.

Criteria to pass this project course

1. Students must attempt at least three different models to address the problem, as relying on a single model will not be acceptable. Failure to meet this requirement will result in the MLP project being disqualified.
2. Additionally, students are expected to:
1. Conduct thorough Exploratory Data Analysis (EDA) to provide meaningful insights and support their modeling choices.

2. Clearly explain their code, demonstrating a solid understanding of its functionality and purpose.
3. Projects that do not meet these standards will not be considered for evaluation.
4. Score cutoff in Kaggle for the final submission to be eligible for viva: 0.80
5. **Passing the project: Viva score >= 25/50 AND Total project score >= 50/100.**

The formula for computing Total Score (out of 100) is as follows:

Final marks = NB + M + S + V

Where,

NB = score of your notebook out of 10 marks, this is assigned by the viva examiner - EDA (4 marks), Model Training and Selection (3 marks), Comparison of different models (4 marks)

M : 5 marks will be awarded to those who successfully complete all milestones.

S: Marks for the ML model, 5 marks awarded to all those who cross the cutoff + *Score* obtained in Kaggle scaled from 0 to 30; Total 35 marks.

$$Score = 5 + 30 \times \frac{Your\ best\ score - cutoff\ score}{Top\ leaderboard\ score - cutoff\ score}$$

V: Marks in Viva Voce, out of 50 marks.

Final marks = NB + M + S + V

Kaggle Competition Link

Following is the link to the competition: https://www.kaggle.com/t/29651a586efd4d448941ba6fa5561e32

The effective leaderboard of the competition is the following:

Click here

The effective leaderboard is computed by only those submissions made via ***FinalNotebook*** of students.

Practice Kaggle Competitions

The following link has good practice Kaggle competitions:

Kaggle Playground Competitions

Project problem type

You will be working on a regression/classification problem.

Rules

1. The MLP project will be conducted on the Kaggle platform.
2. Only students who register for the MLP Project during the course registration window can participate in this project.
3. The link to the project on the Kaggle platform is https://www.kaggle.com/t/29651a586efd4d448941ba6fa5561e32
4. Each student will participate individually only. This is not a group project.
5. No external data can be used to train your model, if found otherwise, your project will be discarded.
6. Do not use any accelerators available in the Kaggle platform. Use a CPU only environment to execute your notebooks, which is by default for every notebook.
7. You are supposed to keep your project code in private mode. You will be disqualified if your code is found public.
8. **Plagiarism is a serious offense. You are suggested to refer to the student handbook to acquaint yourself with the detailed consequences of plagiarism.**

Start making submissions

Once you have joined the competition on Kaggle you can create a new notebook, start submitting ML code on Kaggle within the notebook and see what score you get using it.

You can make a maximum of 5 submissions every day. If you don't make a submission on a day, the 5 chances will not be carried forward the next day.

You can change the models and see if the score keeps improving.

Evaluation of the ML code that is submitted in Kaggle

Watch this video for more information on how to create and manage your notebook.- **Demo video link**

1. Create a new notebook inside the competition code tab, referred to as ***FinalNotebook*** hence onwards.
2. This will be submitted as your final work and will have your best model for evaluation.
3. Name your ***FinalNotebook*** as **"YourRollNo-notebook-t12026",** i.e. your roll no, followed by hyphen/dash("-"), followed by**"notebook-t12026".** All characters should be in lowercase letters. If your roll no is 21F10001234, then your notebook name (i.e. ***FinalNotebook***) should be **21f10001234-notebook-t12026.**

4. Share your ***FinalNotebook*** with the admin of the competition i.e. Kaggle username **iitmbscs2008p**.
5. Do not change the name of this notebook. You can create many other notebooks for experimenting with different models and approaches.
6. If you share a notebook with any other name, that notebook will not be counted towards your project.
7. Only submissions made from the ***Final Notebook*** will be considered for grading.

Your first submission:

8. Make your first submission from ***FinalNotebook***. You can use any ML model (e.g. dummy model) for your first submission. We will check from our side to see if we have received it.
9. Submissions made by explicitly uploading ".csv", ".zip" or any other file, will not be counted towards your project.

Milestones:

10. There will be some milestones in the project. Use milestones as intermediate goals for steadily making progress.
11. **Submitting at least 2 out of 5 milestones is mandatory to complete the project. There will be one mark awarded for each milestone completion.**

Registration Status

You can check your registration status from the following link: Registration Status Sheet

If your registration is not complete, it is likely due to the following issue:

1. **is NotebookName correct?** : Your notebook name is not according to the mentioned format
2. **Notebook Shared with iitmbscs2008p?** : You haven't given view access to the competition host (iittmbscs2008p)
3. **Leaderboard score missing**: Your Kaggle username is not present on the leaderboard because you haven't made any successful prediction submissions using your notebook

The sheet is updated every 72 hours. If you have followed all the instructions correctly, please wait for 72 hours before reporting in case of a discrepancy in the Registration sheet.

Milestones and activities:

Following the milestones and completing the corresponding assignments is mandatory to complete the project. The questions for the milestones will be shared via a form in the course portal. The milestone is considered submitted only when you receive the response copy to your email ID.

	Deadline	**Description**
Milestone 1	Feb 15, 2026 Sunday	• Data loading • Exploratory Data Analysis • Data visualization • Statistical analysis • Train-Validation Split • Data Cleaning/ Preprocessing • Building a baseline model.
Milestone 2	Feb 22, 2026 Sunday	1. Linear Models 2. Stochastic Gradient Descent Tune the hyperparameters of the model.
Milestone 3	Mar 1, 2026 Sunday	Dimensionality Reduction Feature Selection Retrain the models Naive Bayes Nearest Neighbors Support Vector Machines
Milestone 4	Mar 8, 2026 Sunday	Ensemble (Bagging, Boosting, Stacking) Multi-layer Perceptron
Milestone 5	Mar 15, 2026 Sunday	Repeat the ML end-to-end process again with insights into the dataset.

Last Submission	15-03-2026 Sunday	Last submission for the project.

There is an assignment in the corresponding week for each of the above milestones in your project in the Seek portal.

Important note: The MLP Project milestones are not in sync with the MLP course. Ideally, the project should be taken after completing the course. The option of taking the MLP course and MLP project together i.e. in the same term, is open for students already having experience with ML (e.g. professional data scientists, data analysts). If you are new to ML, you should complete the MLP course in one term and the MLP project in a later term.

Making a submission

Watch this video for more information on how to create and manage your notebook.- **Notebook Submission Demo link**

1. Create a Kaggle notebook.
2. The code to read and load the data will be already present in the notebook.
3. Your notebook should produce a 'submission.csv' file containing predictions for test samples. Your newly created submission file should be visible in the 'output' tab.
4. Submit the submission file. (for details, refer the video link given)
5. Make sure the notebook name, its version number, and the submitted file name are as desired.
6. Always make a submission from your Kaggle notebook.

When will the project submission be discarded?

1. Submitting a different Kaggle notebook link in Competition Registration Form, and making submissions with another.
2. The python notebook containing the final ML code not being submitted on the Kaggle portal - ***FinalNotebook*** not shared with the competition host.
3. Any student found to plagiarize from the internet or fellow student or any other source will fail this project and will have to repeat this project.
4. Any student found to submit code with any machine learning libraries (such as TensorFlow, PyTorch, NLTK, word2vec, textblob etc) other than those mentioned below will fail this project.

Plagiarism in project

The project will be considered complete only once the final grade for the project course is published on the student dashboard.

Any inappropriate conduct during the project period may be treated as malpractice. This includes, but is not limited to the following:

- Sharing code snippets, notebooks, or any project-related material with any other students. This includes students not registered for the project course.
- Submitting work that is wholly or partially copied from another student
- Allowing another individual to complete any portion of the project on one's behalf.
- Using solutions, code repositories, or notebooks from external sources
- Reproducing project outputs that are substantially similar to another student's work, even if variable names or minor details are altered.

Refer to the student handbook to acquaint yourself with the detailed consequences of plagiarism

Libraries allowed

Only the following libraries will be allowed to be used in the competition:

1. NumPy
2. Pandas
3. Matplotlib
4. Scikit-learn
5. XGBoost
6. Seaborn
7. Imblearn
8. SciPy
9. Pickle
10. regex
11. Lightgbm
12. Plotly
13. statsmodel

14. wordcloud

Please note these libraries are already installed in your Kaggle notebook. Python in-built libraries can also be used. **Using libraries like TensorFlow, PyTorch, NLTK, AutoML frameworks, spaCy, word2vec, textblob etc is not allowed.**

Eligibility to schedule and attend the viva:

You can keep making submissions in the contest every day.

Mandatory condition 1 to be eligible for viva: Your submissions must be ONLY from the Kaggle notebook submitted in Competition Registration Form.

Mandatory condition 2 to be eligible for viva: crossing the score cutoff.

Mandatory condition 3 to be eligible for viva - the submitted ***FinalNotebook*** containing your latest ML model, when executed, must cross the cutoff.

Mandatory condition 4 to be eligible for viva - The Python ***FinalNotebook*** should pass the plagiarism checker.

Where to check Viva eligibility?

You can check if you are eligible for the viva exam from the following document: Viva Eligibility Sheet

We will add - user name status, First/sample submission status, notebook name status, team name status, effective leaderboard rank, score, NB, V, T, P/F

The above information will be updated every Monday EOD after publishing initially.

The cutoff score for the t12026 term Kaggle contest is **0.80**

Level 1 Viva

The Level-1 Viva is designed to ensure that each participant in the Kaggle Project has put in a significant amount of effort in their MLP Project. This ensures that only participants who have shown dedication and effort in their Project work are eligible for the Level-2 Viva. **Please note that completing all checklists is mandatory.**

Following is the checklist for Level 1 viva:

1. The dataset is appropriately loaded and stored into corresponding variables.
2. Exploratory data analysis
1. Visualizing key statistics and relationships in the data
2. Correctly identify the feature types.

3. Detecting missing data and imputation, if required.
4. The dataset is appropriately preprocessed,
1. scaling numerical features.
2. encoding categorical features.
5. Appropriate usage of pipelines if any.
6. Feature engineering/extraction.
7. Hyperparameter tuning of the model to optimize its performance.
8. Code should be clean, well-structured, and appropriately commented.
9. Highlight important ideas learned from the dataset and/or model.
10. Compare at least 3 best models of all the models experimented, analyze their results, and provide insights on the model's performance.
11. Make submission with the model with the best score on training data.

Note:-

Performing exploratory data analysis (EDA) and making predictions using a minimum of three models is a mandatory requirement. Additionally, achieving a minimum score of Will be updated soon is necessary to successfully pass the Level 1 viva examination.

Repeat L1- Viva

If a student completes less than 50% of the checklist (excluding mandatory checkpoints), they will get a chance to reappear. However, if they complete between 50% and 70%, they will not be allowed to reappear and will be awarded a grade based on their checklist performance: D for 50% to 60% completion and E for 60% to 70% completion.

Level 2 Viva

How to prepare for Level 2 Viva Voce Examination

Kaggle notebook: You are expected to clearly explain the code you have written and logically describe your choice of data preprocessing steps, modeling, and inspection of underfitting/overfitting.

The viva examiner may ask you to briefly (say in 5 minutes or less) walk through your Kaggle notebook.

The viva examiner may ask questions from any course you have taken for your DS diploma. The Kaggle competition may belong to only, say classification, yet the

examiner can ask questions from regression and unsupervised machine learning, i.e. **all topics you have learned in courses till MLP, e.g. MLF, MLT, statistics etc.**

Only using an API (e.g. DecisionTreeClassifier, SVC, etc.) is not sufficient, you should also know how it works internally.

For every machine learning model that you have learnt and/or used in your Kaggle notebook, you must know and be able to explain its 5 components, i.e. data, model, loss function, optimization procedure/training and evaluation.

You may be asked to write code from scratch or make changes to your existing code during your viva exam.

Level 2 Viva marking scheme

For level 2 viva, students are expected to be well-versed with the following details of the components mentioned in the subsequent table:

1. Conceptual understanding

2. Mathematical details

3. Practical aspect

4. Programming

Repeat Viva-voce Examination

In case a student scores between 15-24/50 marks, they will not pass the viva but they will get another chance to appear for the viva examination. If they are not able to clear the viva exam on the second attempt also, they will have to repeat the MLP Project in the next term.

If a student scores <15/50 in the viva examination, they will not get a second attempt for the viva examination. They have to register and pay the fees, repeat the entire project the next term and become eligible to attend the viva. It is seen that such students need more practice and time to study the fundamentals again.

FAQs

Que: Do I have to take the MLP course and the MLP Project in the same term?

Ans: No, you may take the MLP course in one term and the project in the next.

Que: have taken the MLP course and the MLP Project in the same term. However I find that the milestones cover topics which were not covered in the MLP course before the corresponding (milestones') deadlines. How can I complete the project then?

Ans: If you are taking an MLP course and the MLP project in the same term, then you will have to be proactive and learn the MLP topics (from YT playlists etc.). You will have to

follow the pace of the milestones in the project. If you find this to be too hectic, you may complete the MLP course first before taking the project.

Que: What is the link to the MLP course playlist on youtube ?

Ans: https://youtube.com/playlist?list=PLZ2ps__7DhBb3ovNQam2c_WcGeiW9OhQo

Que: Can I take my viva before the project's deadline?

Ans: Yes, if you are confident that you have successfully completed the project and your Kaggle notebook is ready (refer to the viva section), and if you are comfortable taking the viva before the project deadline is over, you can. However, grades for your 'score' component will be computed only after the deadline is over. That means you can still make submissions to improve your rank and score after the viva. The final grade will be computed only after you have taken your viva voce exam and the project's deadline is over.

Que: Can I finish the project this term and take the viva next term?

Ans: No, as per the recent committee decision students have to complete the project and viva in the same term. Every term the problem statement will change.

Que: How can I know that I have followed the project registration steps properly and my registration is approved?

Ans: You can track your status in the google sheet provided. You will be given week 2 as a buffer in case information is incorrectly provided in Assignment 1.

Que: When can I expect my final grade for this project?

Ans: 2 days after your viva examination is complete.

Que: How should I approach my project?

Ans: One way to start your project is to quickly build a baseline model. Then keep improving your pipeline for the entire project. You will have to repeat the end-to-end ML (video from MLP Week 1) process multiple times.

Que: I have many models for the dataset, but I am stuck and not able to improve my score.

Ans: You may try hyperparameter tuning and feature engineering. Look at the samples your model is making incorrect predictions, try to learn why that is, and improve your model.

Que: My rank and score are much better than what is displayed in the effective leaderboard?

Ans: This could be possible if you have made submissions via a Kaggle notebook as well as csv/zip files, and your csv/zip files are getting better score and rank. However, while computing the effective leaderboard, your submissions via csv/zip files will be ignored.

Que: What is the link to the MLP Project t12026 term guidelines document?

Ans: You are reading exactly the same! Following is the link to this document: MLP Project t12026 Guidelines

Once the project deadline is over

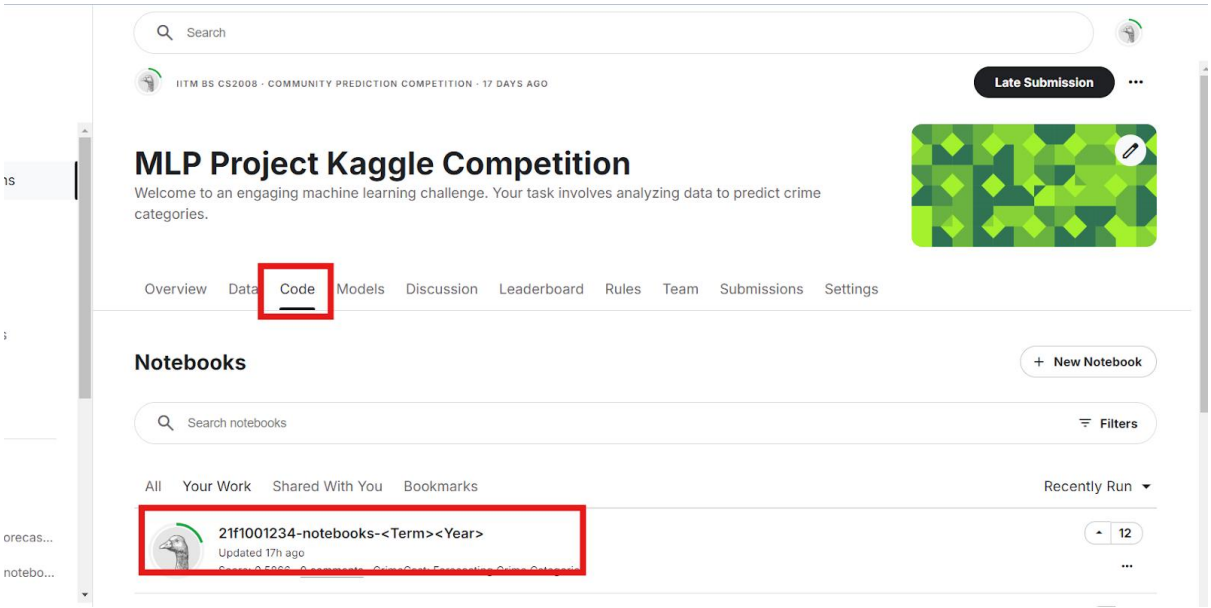
Upload your Kaggle notebooks to GitHub once the project's deadline is over. This will serve to build your profile.

Tentative Viva Schedule

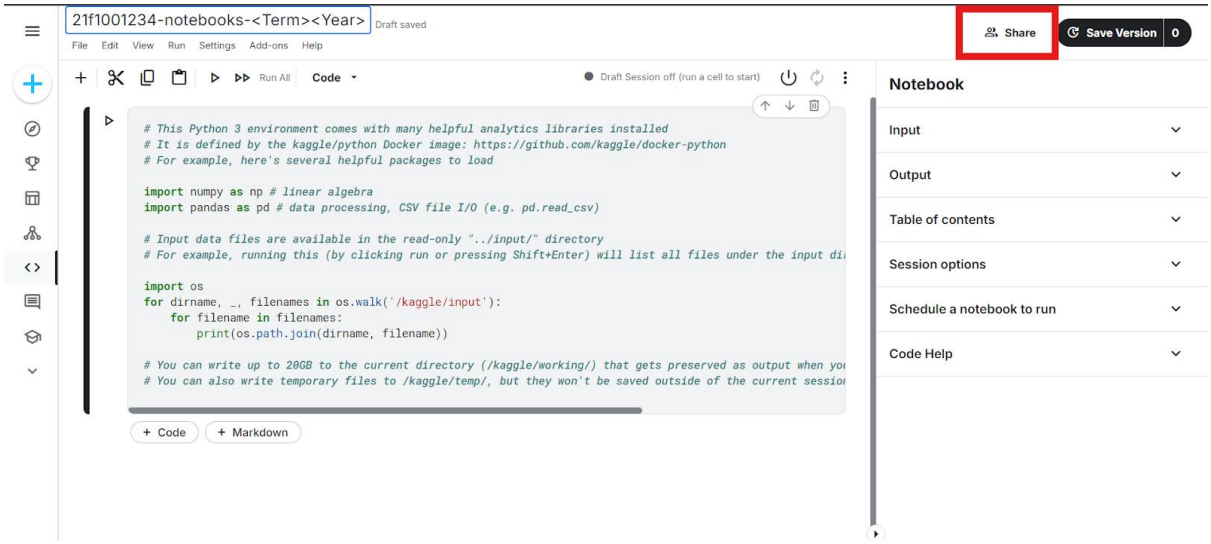
Viva Planner for MLP Jan 2026 term: Viva planner

How to make Admin(iitmbscs2008p) a collaborator to your competition Kaggle Notebook

1. Go to the competition page -> Code-> Your Work. Where you will find Kaggle notebooks that you have created in the competition.

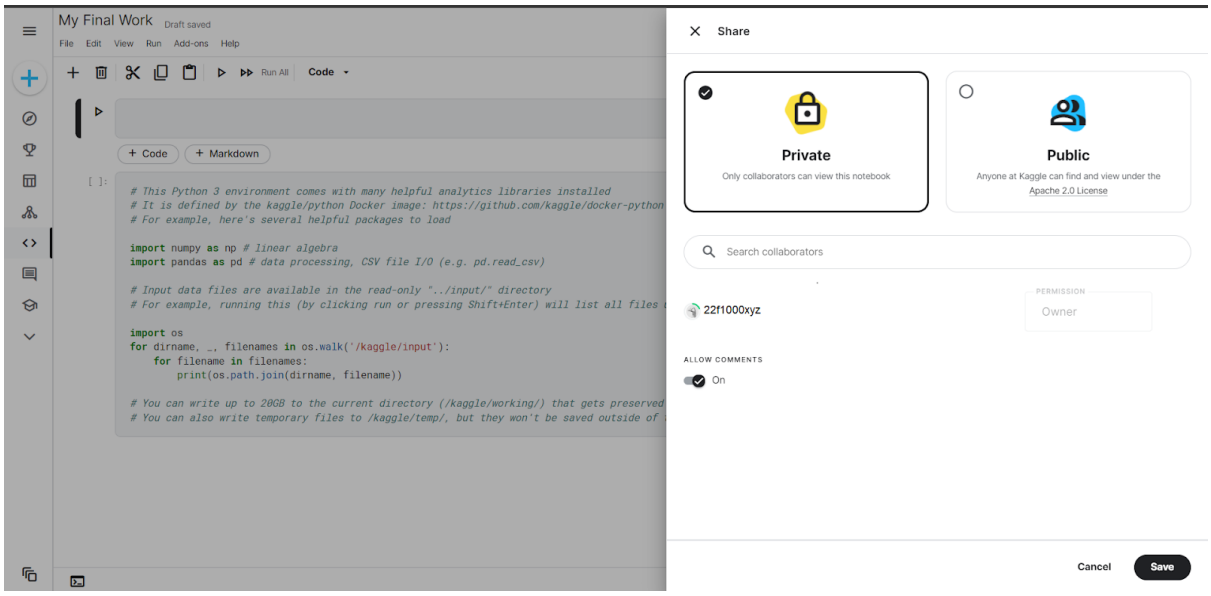


2. Click on the share button on the right top of the page.



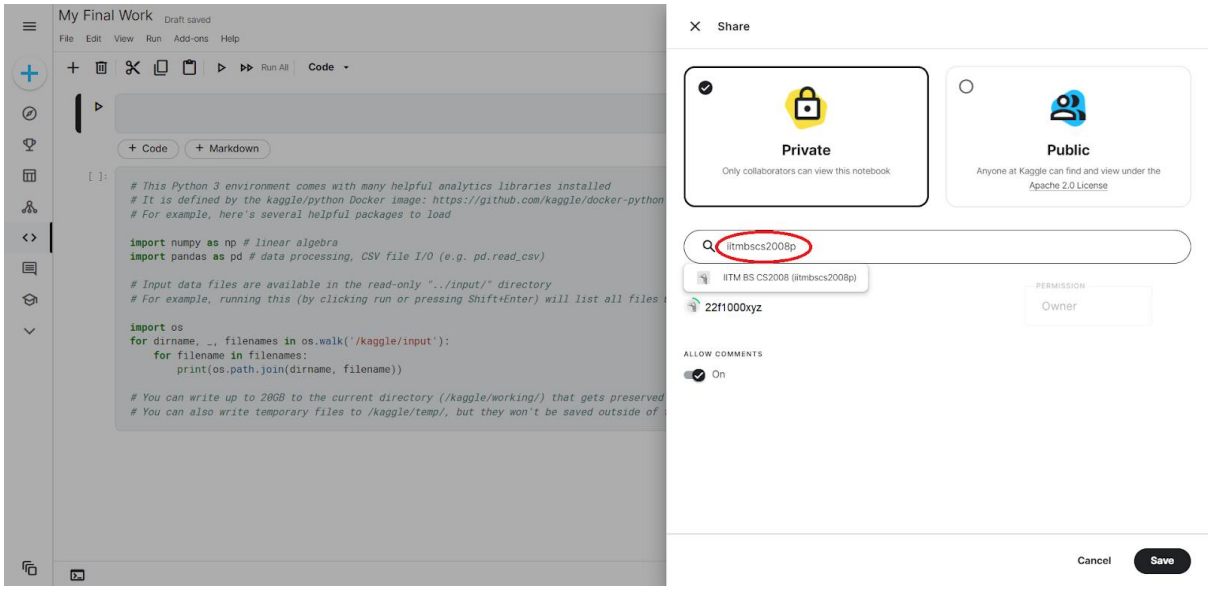
3. The window will pop up. Choose **Private** (Only collaborators can view this notebook)

(**Note:** don't keep it public because you will be offensible if your code is found plagiarized**)**

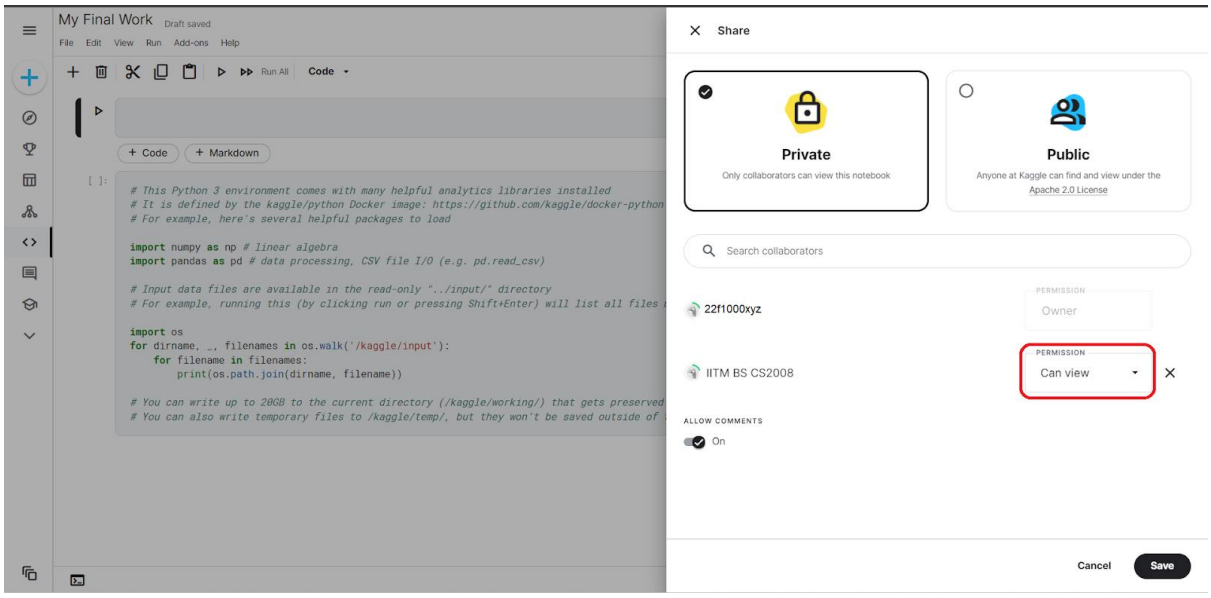


4. Click "search Collaborators". Enter the admin's Kaggle username, which is "iitmbscs2008p" and choose **"iitmbscs2008p".**

(**Note:** Don't add other collaborators because you will be offensible if your code is found plagiarized. Don't make your notebook public ever**)**

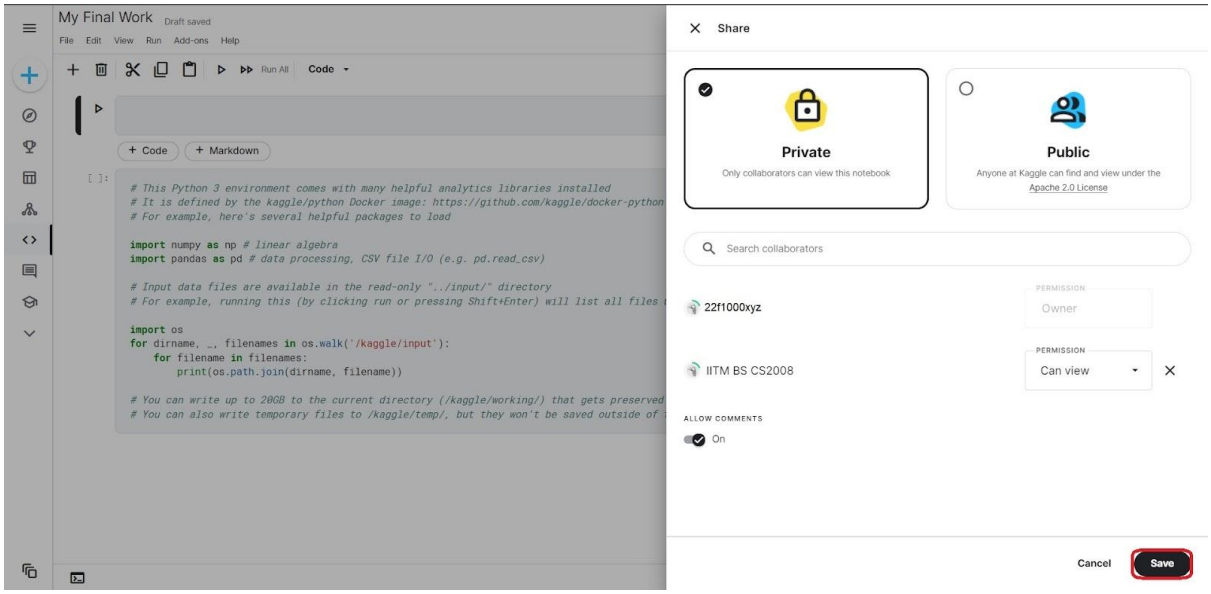


5. Give “can view” access to the display name “**iitmbscs2008p**”.



6. Allow comments: **On**

7. The screen will look like the following: click the “Save” button.



8. Once shared, you can proceed to register for the ML Viva Voce exam.